SOIL EROSION AND CONSERVATION · NOTE GC01

Gully and Ravine Lands — Drainage-Line Treatment, Extent, Land Capability and Reclamation Planning

Course topics: SWCE4.5 L13, SWCE8.15 L6, SWCE-505 T3

Drainage-line treatment

- **Drainage lines** are the outlet routes of a watershed's excess runoff — drainage channels, gullies, ravines, torrents, natural waterways, streams and rivers. They remove excess water from the surface and the root zone, maintain ecological balance and guard against environmental degradation.
- Uncontrolled runoff, steep slopes, high velocities and heavy sediment loads degrade them, forming gullies, degrading land and damaging property and crops.
- **Biological measures** (preferred): vegetative barriers, grass and tree planting — thick, deep-rooted grasses protect drainage lines economically, eco-friendly and permanently. **Mechanical measures** (when vegetation is not enough): check dams, diversion bunds, terracing. Gullies and ravines usually need **integrated** mechanical + vegetative treatment.

Term	Meaning
Gully	A small, elongated eroded depression, usually formed after a rainstorm — linear or branching
Gully erosion	Soil removal by excessively concentrated running water, cutting deep, narrow channels
Incipient erosion	The early stage of gully formation
Fluvial erosion	Erosion by running water
Ravine	A deep, wide gorge cut by running water — larger than a gully, smaller than a canyon
Canyon	A deep valley with steep, rugged slopes and a stream at the bottom — very wide top (can be several km), narrow bottom
Gorge	A narrow opening between hills or mountains

Extent in India

State	Gully and ravine land (ha)
Madhya Pradesh	6,00,000
Bihar	6,00,000
Rajasthan	3,40,000
Gujarat	2,80,000
Uttar Pradesh	1,23,000
Punjab & Haryana	1,20,000
West Bengal	1,04,000
Tamil Nadu	60,000
Maharashtra	20,000
Total	**≈ 22,47,000**

The state figures add up to the total. They are as given in the notes; other surveys report different state-wise figures, so quote the source and year.

- **Causes:** natural — gently sloping land, sandy and silty soils, intense rain; human — improper land use, overgrazing, removal of vegetation and faulty farming.
- **Effects:** loss of farmland, shortage of food and fodder, displacement of villages, lower water table, reservoir siltation and socio-economic problems.

Land capability classification for ravine lands

Based mainly on land slope, gully slope and distance from the gully rim.

Land form	Slope	Distance from gully rim	Class
Table lands	0–1%	Beyond 60 m	I
Table lands	1–3%	Beyond 60 m	II
Table lands and wide humps between gullies	0–3%	6–60 m	III
Table lands	3–5%	Beyond 6 m	III
Table lands	5–10%	Beyond 6 m	III
Table lands	10–15%	Beyond 6 m	IV
Marginal land near the gully rim	0–15%	Within 6 m	VI
Table land near the gully rim	15–25%	Within 6 m	VI
Very small to medium gullies (G1, G2, G3)	—	—	VI
Deep and narrow gullies	—	—	VII

Planning gully control

- Use land according to its capability: cultivate level and moderately sloping land; convert steep, eroded land to permanent vegetation; plant natural waterways with close-growing perennials.

- Gully control may cost more than the land is worth but is justified to protect adjacent land, downstream areas, reservoirs and buildings.
- **Best use of ravine land:** permanent grasses and trees — essential for deep and narrow ravines.

1. Survey and map.
2. Classify land capability.
3. Plan vegetative measures and close the area to grazing and biotic interference.
4. Check the advance of gully heads.
5. Plug gullies.
6. Stabilise gully heads.
7. Reclaim land for cultivation.
8. Reclaim deep gullies.

Survey

- A topographic survey serves land-use classification and planning. Map gully rims and beds, field boundaries, watercourses and vertical gully faces: establish bench marks, survey, and prepare a base map.
- **Scales:** base map 1 : 1200; planning maps 1 : 4000 or 1 : 8000. Use a plane table with telescopic alidade where a dumpy-level survey is difficult.

Closure to grazing

Restricting grazing and other biotic interference increases useful grasses, improves tree growth and reduces erosion.

Reclamation by gully size

Gullies	Treatment
Small	Clearing, minor levelling, diversion bunds at horizontal intervals and grass ramps for runoff disposal — then cultivation is possible
Medium (and ravines)	Clear and level the bed, build a series of check dams, terrace the side slopes: bench terracing on moderately steep slopes; grass on terrace outlets and earthen check dams; store sediment-free runoff in farm ponds for irrigation and domestic use
Deep, narrow (> 9 m)	Permanent vegetation — fast-growing trees and grasses — with afforestation, gully plugging and grass planting

Gully plugging is a series of check dams (temporary, semi-permanent or permanent) across a gully that slows flow, controls erosion, stabilises the bed and makes silt settle upstream — preventing further gullying, improving moisture and stabilising the drainage line.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ICAR–Indian Institute of Soil and Water Conservation, Research Centre, Agra / Vasad (ravine reclamation). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. A ravine is

(a) smaller than a gully
(b) a deep, wide gorge larger than a gully
(c) a canyon
(d) a rill

Q2. Total gully and ravine land in India (as in the notes) is about

(a) 2.25 lakh ha
(b) 22.47 lakh ha
(c) 224.7 lakh ha
(d) 67 lakh ha

Q3. Table land of 0–1% slope beyond 60 m from the gully rim is class

(a) I
(b) II
(c) III
(d) IV

Q4. Marginal land within 6 m of the gully rim is class

(a) III
(b) IV
(c) VI
(d) VIII

Q5. Deep and narrow gullies fall in class

(a) IV
(b) V
(c) VI
(d) VII

Q6. Base maps of gullied areas are prepared at a scale of

(a) 1 : 1200
(b) 1 : 15,000
(c) 1 : 50,000
(d) 1 : 250,000

Q7. Best land use for deep ravines is

(a) intensive cropping
(b) permanent grasses and trees
(c) paddy
(d) orchards with clean cultivation

Answers

Q1 — (b) a deep, wide gorge larger than a gully
Q2 — (b) 22.47 lakh ha
Q3 — (a) I
Q4 — (c) VI
Q5 — (d) VII
Q6 — (a) 1 : 1200
Q7 — (b) permanent grasses and trees